



Analysis of light field and flow field of vortex rods tube photobioreactor based on field synergy principle

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ABSTRACT

A new static mixer called the vortex rod static mixer was introduced into the photobioreactor. This study focused on the vortex rod tubular reactor as the subject of investigation. The effects of different incident light field intensities ($I_0 = 375 \mu\text{mol}/(\text{m}^2\cdot\text{s})$, $800 \mu\text{mol}/(\text{m}^2\cdot\text{s})$, and $1200 \mu\text{mol}/(\text{m}^2\cdot\text{s})$) on reactor efficiency were simulated using the CFD simulation method, considering variations in vortex rod arrangement (60° , 90° , and 120°) and tilt angle ($\alpha = 27^\circ$, 36° , 45° , 54° , and 63°). In the SST turbulence simulation flow field results, the movement process of algae cells in the tube was simulated using the particle tracking module. By taking into account the average frequency of the light-dark cycle, the enhancement efficiency of the light-dark cycle, and the synergy angle of the light gradient and velocity field, the following results were obtained: the optimum tilt angle for the vortex rod is $\alpha = 45^\circ$. For incident light field intensities of $I_0 = 375 \mu\text{mol}/(\text{m}^2\cdot\text{s})$ and $800 \mu\text{mol}/(\text{m}^2\cdot\text{s})$, the 60° arrangement yielded the best results, resulting in a 54.491 % and 66.258 % increase in cycle frequency, an 18.747 % and 22.795 % enhancement in efficiency, and an average $|\cos\beta| = 0.68055$. For an incident light field intensity of $I_0 = 1200 \mu\text{mol}/(\text{m}^2\cdot\text{s})$, the 90° arrangement proved to be the most effective, resulting in a 57.580 % increase in cycle frequency, an 18.058 % enhancement in efficiency, and an average $|\cos\beta| = 0.68975$.

1. Introduction

Algae, contributing over half of the Earth's oxygen [1], present a promising solution to food and protein shortages due to climate change and population growth [2]. Utilizing solar energy for photosynthesis to fix CO₂ is widely studied for its environmental sustainability and cost-effectiveness [3]. Microalgae, with high photosynthetic efficiency and rapid growth rates, can be cultivated for high-concentration product generation through photoautotrophic culture [4,5], offering an eco-friendly CO₂ immobilization option [6]. They are also rich in valuable bioactive compounds [7]. Microalgae possess high photosynthetic efficiency, rapid growth rates, and significant oil content. They are abundant in proteins, lipids, pigments, and other active substances, making them vital resources in various fields [8–11].

To overcome the challenges of microalgae utilization, closed photobioreactors like tubular PBRs are employed, providing stable culture conditions, a larger illuminated area, and high-density culture [12]. Tubular photobioreactors are favored for outdoor large-scale cultivation [13,14].

Light plays a significant role in microalgae growth, and controlling

the light-dark cycle can enhance photosynthetic efficiency. When the light intensity is below the light saturation point, the photosynthetic rate of microalgae increases with increasing light intensity [15]. Beyond the photoinhibition point, the excessive light intensity can damage the enzymes responsible for photosynthesis in algal cells [16], thereby inhibiting photosynthesis. Increasing the frequency of the light-dark cycle promotes the growth of algae cells [17,18]. Effective mixing in the reactor is essential for facilitating the light-dark cycle and improving light energy utilization.

To improve fluid mixing in the reactor, two approaches are used: dynamic mixing and static mixing. Static mixing employs stationary plug-ins in the pipeline, eliminating mechanical mixing and offering a simple structure and low maintenance costs. Studies have investigated the effects of static mixers in photobioreactors on mass transfer and the light-dark cycle of algae cells within the tube.

Ugwu [19,20] examined the impact of four different static mixers on mass transfer performance in tubular photobioreactors. The results showed improved mass transfer compared to a plain tube. Further research revealed that the addition of static mixers facilitated the movement of algae cells between light and dark areas, thereby

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increasing the biological yield of algal liquid. Cheng [21] introduced a novel type of cross-half-moon porous static mixer. Compared to a smooth tube reactor, the average velocity along the light direction increased by nearly 1000 times, and the turbulent kinetic energy increased by 1.3 times. At inlet velocities of 0.3 m/s and 0.4 m/s, the light-dark cycle frequency increased by 85.21 % and 95.77 %, respectively.

Qin [22] investigated the arrangement of static Kenics mixers in tubular photobioreactors and found that the continuous arrangement of static mixer units did not lead to significant improvement. The light-dark cycle frequency increased slightly, while the pressure drop significantly increased. Interval arrangement proved to be more effective. Qin et al. [23] introduced discrete oblique ribs into the tubular photobioreactor. CFD simulation results demonstrated the formation of multiple longitudinal vortices in the tube's cross-section, promoting the flow of algae cells in the direction of the light gradient and increasing the frequency of the light-dark cycle.

Zhang et al. [24] utilized the CFD method to study the flow field characteristics in a tubular photobioreactor with an embedded spiral mixer. They obtained the optimal size of the spiral mixer and subsequently confirmed through experiments that the introduction of spiral mixers increased the biomass yield of algae cells by 37.26 %.

During the process of fluid-enhanced heat transfer, some researchers have proposed the utilization of vortex rod design to achieve optimal velocity fields based on the concept of field synergy. This approach aims to enhance heat transfer performance while maintaining a fixed pressure drop [25]. Insights from previous studies in other fields have inspired the design of novel photobioreactors. Specifically, a simple structure can be employed to generate longitudinal vortices and enhance the light-dark cycle frequency of algal liquid.

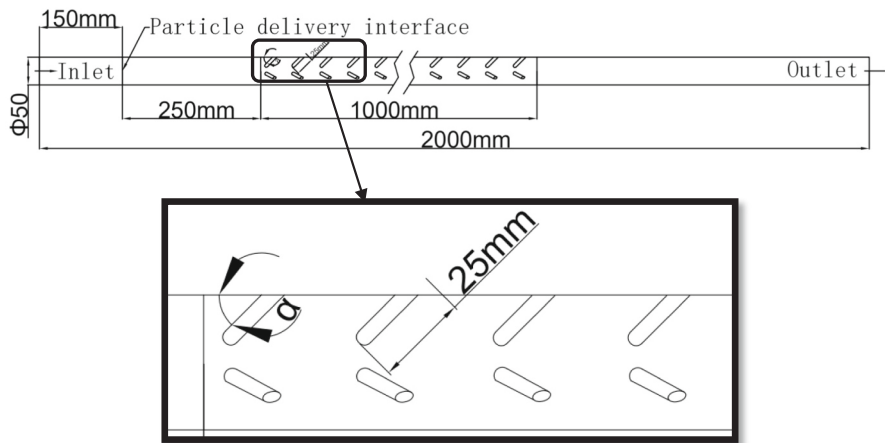
In this study, a vortex-rod tubular photobioreactor is proposed, comprising a series of plug-in vortex rods. To achieve this design, evenly spaced openings are created on the tube wall during the production of

the tubular photobioreactor. Transparent vortex rod components with appropriate working angles are then inserted into these openings, and proper sealing is ensured using gaskets and sealing rings. This arrangement allows for the generation of multiple longitudinal vortex flows using the vortex rods, effectively enhancing convection between the tube wall and the core area. Considering that in intensive culture, the light region is typically located near the tube wall, while the dark region encompasses the core area of the tube, this unique flow structure can significantly improve the light-dark cycle.

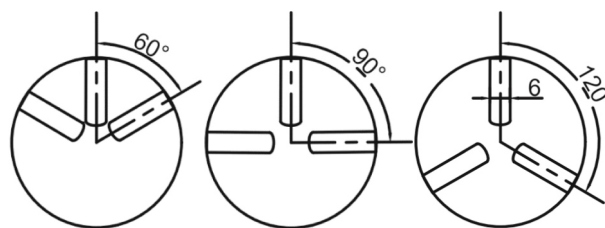
2. Model

2.1. Geometric model

The calculation domain and model size employed in this study are presented in Fig. 1(a). Since the inner diameter of outdoor large-scale tubular photobioreactors is generally 50 mm, the enlargement of the tubular photobioreactor size is limited by the tube diameter. If the diameter is too large, light attenuation occurs, and the algae cells in the central region of the tube cannot receive sufficient light, which restricts their growth. Therefore, the inner tube diameter is set to 50 mm. Considering both computational accuracy and cost, the length of the calculation domain is set to 2000 mm, and the static mixer region extends 1000 mm. Previous studies have shown that a small spacing between vortex bar units can result in better fluid mixing between the wall and core regions [25]. A spacing equal to the tube diameter is considered a good value. To enhance the disturbance of the liquid inside the tube and reduce the dead zone in the core flow, the vortex rod's length is set at 25 mm, considering its potential for aeration expansion, while the rod diameter is chosen as 3 mm. Among these parameters, ' α ' represents the inclination angle of the vortex rod, which is one of the optimization parameters examined in this paper. Specifically, the investigated values of ' α ' include 0° (representing a light tube), 27°, 36°, 45°, 54°, and 63°.



(a) reactor size diagram



(b) Angle layout diagram

Fig. 1. Reactor structure model diagram.

Fig. 1(b) illustrates the arrangement angle of the vortex rods within the tube along the flow direction, which is another optimization parameter explored. The angles of 60°, 90°, and 120° are chosen for investigation.

To ensure calculation stability, a local unidirectional condition is imposed on the outlet boundary to minimize the impact of the fluid inlet and promote fully developed turbulence. The inlet is configured to facilitate a fully developed flow, with a smooth tube section extending 150 mm to prevent backflow at the outlet boundary and enhance simulation accuracy. Moreover, a 250 mm extension section is included at the particle release interface to ensure complete fluid development during particle release. At the outlet of the mixer, a 600 mm extension is incorporated to prevent backflow. The outlet condition is set as a pressure outlet condition.

2.2. Turbulence model

Algae flow in a tubular photobioreactor is commonly considered turbulent flow [26]. Chlorella, which is spherical with a density close to that of water and a cell diameter of approximately 5 μm, is much smaller than the Kolmogorov scale ($\eta \sim \text{LRe}^{-0.75} = 50 \mu\text{m}$) [27]. Therefore, in this study, it is assumed that the flow properties of the algae solution are similar to water, neglecting the influence of algae cells on the flow field.

The problem at hand is assumed to be three-dimensional, turbulent, and steady. The following assumptions are made for deriving the governing equations: (1) the vortex rod is rigid, disregarding any vibration or deformation; (2) the fluid is continuous and incompressible; (3) the effects of gravity and viscous heating are neglected. In this paper, the two-equation model, RANS SST model, is employed. Convergence is considered achieved when the residual value is below 10^{-6} . The governing equations are presented as Eqs. (1) and (2).

$$\nabla u = 0 \quad (1)$$

$$\rho \frac{du}{dt} + \rho(u \cdot \nabla)u = \nabla \cdot \left[-pi + \mu(\nabla u + (\nabla u)^T) - \frac{2}{3}\mu(\nabla \cdot u)i \right] + F \quad (2)$$

Turbulent kinetic energy Eq. (3):

$$\rho \frac{dk}{dt} + \rho u \cdot \nabla k = \nabla \cdot [(\mu + \sigma_k \mu_T) \nabla k] + P - \rho \beta_0 \rho \omega k \quad (3)$$

Turbulent kinetic energy ratio dissipation rate Eq. (4):

$$\rho \frac{d\omega}{dt} + \rho u \cdot \nabla \omega = \nabla \cdot [(\mu + \sigma_\omega \mu_T) \nabla \omega] + \frac{\gamma}{\mu_T} \rho P - \rho \beta \omega^2 + 2(1 - f_{v1}) \frac{\sigma_{\omega 2} \rho}{\omega} \nabla k \cdot \nabla \omega \quad (4)$$

2.3. Illumination model

Considering the significant impact of illumination on microalgae growth, the studied vortex photobioreactor was constructed using fully transparent materials (for example, ternary ethylene-propylene rubber). Therefore, only the self-attenuation process of light within the algal suspension was taken into account. Due to the high-density cultivation of microalgae cells in the photobioreactor, the concentration of algal cells is significantly increased. Previous studies have shown that the light attenuation phenomenon occurs during high-concentration microalgae cell culture [28]. In this context, the Cornet model is considered more appropriate than the Lambert-Beer model, particularly when the backlit surface of the tubular reactor becomes nearly opaque in high-density culture. Therefore, the Cornet model is utilized to calculate the light intensity within the reactor. The expression of the Cornet model is as follows:

$$I = \frac{4\alpha_1 I_0}{(1 + \alpha_1)^2 \cdot e^{\alpha_2} - (1 - \alpha_1)^2 \cdot e^{-\alpha_2}} \quad (5)$$

$$L = \sqrt{r^2 - x^2} - y \quad (6)$$

$$\alpha_1 = \sqrt{\frac{E_a}{E_a + E_s}} \quad (7)$$

$$\alpha_2 = (E_a + E_s) \cdot \alpha_1 \cdot C_x \cdot L \quad (8)$$

To ensure the sufficient discrimination of light intensity conditions, $I_0 = 375 \mu\text{mol} / (\text{m}^2 \cdot \text{s})$, $800 \mu\text{mol} / (\text{m}^2 \cdot \text{s})$ and $1200 \mu\text{mol} / (\text{m}^2 \cdot \text{s})$ were selected. When the cell concentration $C_x = 1.3 \text{ g} / \text{L}$, according to the literature [29], the light absorption coefficient $E_a = 0.0014 \text{ m}^2 / \text{g}$ and the light scattering coefficient $E_s = 0.9022 \text{ m}^2 / \text{g}$ were selected.

2.4. particle tracing

The Particle Tracing For Fluid Flow (FPT) module is used to simulate the particle trajectory. The particles follow Newton's second law of motion:

$$\frac{d}{dt} \left(m_p \frac{dq}{dt} \right) = F_t \quad (9)$$

Where m_p is the mass of the particle, q is the position vector of the particle, and F_t is the net force or total force acting on the particle. For a particle sinking in the fluid, the total force is the sum of gravity F_g and drag F_D .

$$F_g = \frac{\rho_p - \rho}{\rho_p} m_p g \quad (10)$$

$$F_D = \frac{1}{\tau_p} m_p (u_p' - u) \quad (11)$$

$$u_p' = u_p + \Delta u_p \quad (12)$$

Particle velocity response time:

$$\tau_p = \frac{4\rho_p d_p^2}{3\mu C_D Re_r} \quad (13)$$

Among them, ρ_p is the density of the particles, ρ is the density of the surrounding fluid, g is the acceleration caused by gravity (about $9.8 \text{ m} / \text{s}^2$ below sea level), μ is the dynamic viscosity of the surrounding fluid, d_p is the particle size, u is the velocity of the surrounding fluid, and v is the velocity of the particles ($v \equiv dq / dt$).

The selection of particle boundary conditions is based on the initial positions, which depend on the flow field velocity in the turbulence module. This approach aims to release particles with a number density proportional to the fluid velocity, ensuring that the particles at the particle release section tend to approximate a well-mixed suspension of microalgae. This differs from the common practice in most literature where particles are placed at fixed points. The particle release method used here accounts for virtual mass and pressure gradient forces. For the handling of particle-wall collisions, a full scattering condition is adopted to simulate the discrete random motion of microalgae on the wall. These settings ensure that the particle motion inside the reactor maximally mimics the movement of microalgae, providing an accurate representation of the microalgae's motion process.

3. Performance indicators

3.1. Statistic of the L/D cycle

The assessment of microalgae culture improvement in a photobioreactor relies significantly on the average light-dark cycle frequency of algal cells. To determine the positional changes of particles within the reactor and calculate the light intensity experienced by the particles at different times, the Eulerian-Lagrangian method is commonly used in current research. The light intensity experienced by the particles is influenced by the critical light intensity, denoted as $I_c = 96.84 \mu\text{mol} /$

($\text{m}^2 \cdot \text{s}$). The region above this critical light intensity is known as the light region, while the other half is referred to as the dark region. By analyzing the duration particles spend in the dark region compared to the light region, the light-dark cycle period of the particle can be determined, allowing for the calculation of the cycle frequency. The particle light-dark cycle period is defined as follows:

$$t_c = t_l + t_d \quad (14)$$

Where t_l is the time that the particle stays in the light region in a cycle, t_d is the time that the particle stays in the dark region. If there is a particle that does not complete the complete time cycle within the statistical time range, to calculate the average value of the particle, the process is regarded as a complete time cycle, and the particle's $t_c = t_l$ or t_d is assumed. For a single particle, the average light-dark cycle period is defined as:

$$t_{c,av}^i = \frac{\sum_{l=1}^n t_c}{n} \quad (15)$$

Among them, i is the particle number, and n is the number of particle cycles. To eliminate the randomness of the particle trajectory, 3000 particles are selected for delivery, so the average light-dark cycle period of the total number of particles can be calculated:

$$t_{c,av} = \frac{\sum_{i=1}^N t_{c,av}^i}{N} \quad (16)$$

N is the total number of particles, and the average light-dark cycle frequency of the particles is obtained:

$$f_{av} = \frac{1}{t_{c,av}} \quad (17)$$

3.2. Enhancement efficiency of the light-dark cycle

While enhancing the average light-dark cycle frequency is essential for optimizing photobioreactors, it is worth noting that there is a trade-off between the cycle frequency and fluid pressure drop in the tube, as demonstrated in studies. Increasing the cycle frequency often leads to a higher pressure drop, which necessitates more pump power and incurs additional operational costs for reactor maintenance. Previous studies have introduced a light-dark cycle efficiency index to evaluate the economic implications of the reactor [23]. The light-dark cycle efficiency (L/D cycle enhancement efficiency) is defined as the dimensionless increment of the L/D cycle frequency divided by the dimensionless increment of energy consumption per unit time. This index is used to characterize the economic advantages or disadvantages of the reactor. The light-dark cycle enhancement efficiency is defined as follows:

$$\mu = \frac{\Delta f_{av} / f_{av,0}}{\Delta E / E_0} \quad (18)$$

Among them, $f_{av,0}$ and E_0 represent the average light-dark cycle frequency and power consumption per unit time of algal cells in the tube without vortex rods. Δf_{av} and ΔE denote the average light-dark cycle frequency and power consumption per unit time of algal cells in the tube after adding vortex rods. In this study, the dimensionless increment of power consumption per unit of time, as introduced by Gómez-Pérez [30], can be represented as $\Delta E / E_0 = (\Delta P - \Delta P_0) / \Delta P_0$. Since the cross-sectional area and inlet flow rate of the smooth tube and the vortex rod reactor are the same in this research, the dimensionless increment of power consumption per unit of time can be considered solely as an expression related to pressure drop:

$$\Delta E / E_0 = \frac{(\Delta P - \Delta P_0)}{\Delta P_0} \quad (19)$$

3.3. light gradient and velocity field synergy angle

In the initial stages, the design of photobioreactors primarily relied on trial and error approaches, leading to slow research progress. To tackle the essential aspects of light and mixing in microalgae growth, researchers have adopted the field synergy theory to evaluate the level of synergistic enhancement between the light field and flow field within the reactor [31]. Furthermore, they have employed the synergy angle parameter to optimize the structure of microalgae photobioreactors. The synergy angle, which characterizes the alignment between the light gradient and velocity field, is defined as follows:

$$|\cos\beta| = \left| \frac{\text{grad}I \cdot U}{|\text{grad}I| \cdot |U|} \right| \quad (20)$$

In the formula, I is the light intensity of the reactor; U is the flow field velocity at the reactor. The closer the value of $|\cos\beta|$ is to 1, the better the synergy between light gradient and velocity field is, and the more favorable it is for microalgae growth.

4. Results and analysis

4.1. Model validation

To validate the feasibility of the fluid model, numerical simulations were conducted using the SST turbulence model in a smooth tube. The calculated results were compared with experimental data [32], as shown in Fig. 2. The numerical results demonstrate good agreement with the experimental correlation, with a relative deviation of less than 4.2 %. This indicates that the numerical method of the fluid model used in this study is reliable.

To validate the feasibility of the light model, this study utilized the Cornet scattering model to simulate the light intensity distribution within the structure of a column photobioreactor as described in reference [33]. The calculated results were compared with the experimental measurements from the literature, as shown in Fig. 3. It can be observed that the simulated results follow the experimental trends and exhibit good agreement numerically, demonstrating the reliability of the numerical method used in the fluid model of this study.

4.2. Light field distribution and flow field characteristics of vortex rod reactor

According to Eq. (5), the distribution map illustrating the intensity of

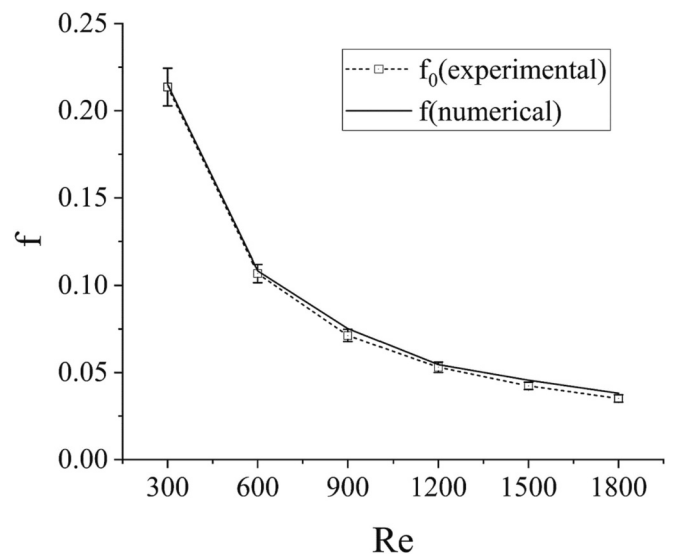


Fig. 2. Comparisons between numerical results and experimental correlations.

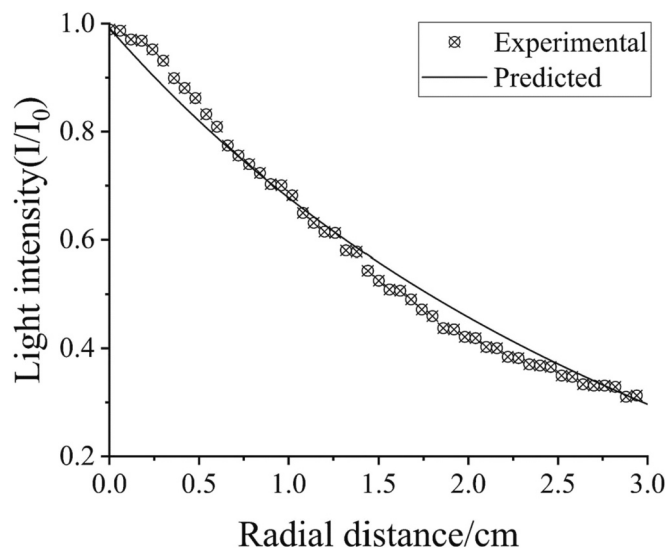


Fig. 3. Comparisons between numerical results and experimental correlations.

the light field across the cross-section of the photobioreactor is obtained, as depicted in Fig. 4. As the incident light field intensifies, the boundary separating the light and dark regions within the tube gradually shifts downward, resulting in an augmented proportion of the illuminated area and a diminished dark area. By increasing the intensity of the incident light, it becomes possible to regulate the distribution ratio between the light and dark regions, thereby enhancing the spatial distribution of the light field within the tube. However, it is important to note that at high intensities of the incident light field, the light intensity within the illuminated area may become excessively strong for the optimal growth of microalgae, potentially impeding their development. Furthermore, excessively high intensities of the incident light field can result in escalated operating costs. Therefore, the enhancement of the incident light intensity must take into consideration multiple factors in order to strike a suitable balance.

The cross-sectional distribution of the flow field within the reactor can be obtained through simulation using CFD software, as depicted in Fig. 5. The simulation is conducted with a fully developed flow inlet velocity of 0.5 m/s and a pressure outlet boundary condition. The results show that the presence of the vortex rod induces the formation of three pairs of vortices within the tube. These vortices are approximately symmetrically distributed along the center of the vortex rod. As the inclination angle of the vortex rod increases, the color of the flow field streamlines becomes more pronounced, indicating a stronger generation of vortices. This highlights the influence of the inclination angle on the size and intensity of the vortex field within the reactor.

It is important to note that in Fig. 5(a), under the 60° distribution, as the tilt angle of the vortex rod gradually increases, the vortices above the vortex rod on both sides diminish and eventually disappear. This suggests that an excessive increase in the tilt angle can disrupt the vortex

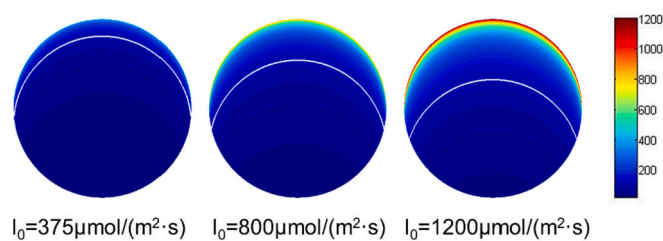


Fig. 4. Illumination field under different algae concentrations and incident light intensities (white line represents the light / dark boundary, i.e., $I_c = 96.84 \mu\text{mol} / (\text{m}^2 \cdot \text{s})$).

structure. Therefore, when designing and optimizing the tilt angle of the vortex rod, it is crucial to select an appropriate angle that aligns with the requirements of the field synergy longitudinal vortex theory.

The arrangement of the vortex rods also significantly affects the flow field within the reactor. In Fig. 5(a), it is observed that when the vortex rods are closely arranged, the lower vortex becomes larger, promoting the light-dark cycle flow of microalgae and preventing the accumulation of microalgae in a specific area, thus avoiding the formation of a stagnant zone. The upper vortex, on the other hand, becomes smaller, resulting in faster movement and a quicker cycle of microalgae in the illuminated zone. In Fig. 5(b), when the vortex rods are distributed at 90°, the size of the vortex field within the tube becomes more uniform. The upper vortex exhibits greater stability, while the lower vortex enhances the exchange of microalgae. Fig. 5(c) demonstrates that when the vortex rods are evenly distributed, the vortex within the tube is symmetrically distributed and stable. However, the lower vortex remains confined primarily to the dark area below, hindering the exchange of microalgae within the tube. Furthermore, it is evident from Fig. 5 that as the arrangement of the vortex rods becomes more dispersed, the disturbance in the center of the vortex symmetry decreases, resulting in poor utilization of light energy by the microalgae cells. Increasing the inclination angle of the vortex rods can significantly improve this situation.

4.3. Effect of angle change on a light-dark cycle

Fig. 6 presents the pressure difference between the inlet and outlet of the photobioreactor for different angle arrangements with inclined vortex rods. The case at 0° represents the pressure difference between the inlet and outlet of a smooth tube without a vortex rod static mixer. The more compact the vortex rod arrangement, the lower the pressure drop at the inlet and outlet of the reactor. However, with an increasing inclination angle of the vortex rod, the pressure drop at the inlet and outlet increases more rapidly. With an even arrangement of the vortex rod, the pressure drop within the reactor tube is higher, but it shows more stable growth as the inclination angle changes.

Fig. 7 illustrates the relationship between the average light-dark cycle frequency and the enhancement efficiency of the light-dark cycle in the reactor under different inclination angles of the vortex rod. Three conditions of the incident light field are considered: $I_0 = 375 \mu\text{mol}/(\text{m}^2 \cdot \text{s})$, $I_0 = 800 \mu\text{mol}/(\text{m}^2 \cdot \text{s})$, and $I_0 = 1200 \mu\text{mol}/(\text{m}^2 \cdot \text{s})$. It can be observed that a more compact arrangement of the vortex rod generally leads to a higher average light-dark cycle frequency and light-dark cycle enhancement efficiency in the reactor, resulting in better performance. However, as the inclination angle of the vortex rod increases, the light-dark cycle enhancement efficiency in the tube decreases more rapidly under the compact arrangement. The impact of light intensity enhancement is more significant on the performance of the photobioreactor with a compact arrangement. As the intensity of the incident light field increases, the average light-dark cycle frequency in the tube decreases at low inclination angles and is better enhanced at high inclination angles. The overall light-dark cycle enhancement efficiency decreases, but the downward trend gradually transitions into a growth trend. When the vortex rod is arranged at 90°, it exhibits improved performance under high light conditions, but the performance of the photobioreactor does not further increase as the light intensity continues to rise from $800 \mu\text{mol}/(\text{m}^2 \cdot \text{s})$ to $1200 \mu\text{mol}/(\text{m}^2 \cdot \text{s})$. The intensity of the incident light field in the photobioreactor with a uniform arrangement of vortex rods has minimal effect, resulting in a low average light-dark cycle frequency and light-dark cycle enhancement efficiency.

In Fig. 7(a), under a light field of $375 \mu\text{mol}/(\text{m}^2 \cdot \text{s})$, the reactor's average light-dark cycle frequency reaches a peak with varying inclination angles of the vortex rod. The maximum value of the average light-dark cycle frequency ($f_{av} = 0.8047 \text{ Hz}$) is achieved when the inclination angle (α) is 54° with a 60° arrangement. In the 90° and 120° arrangements, $\alpha = 45^\circ$ and 54° yield similar average light-dark cycle

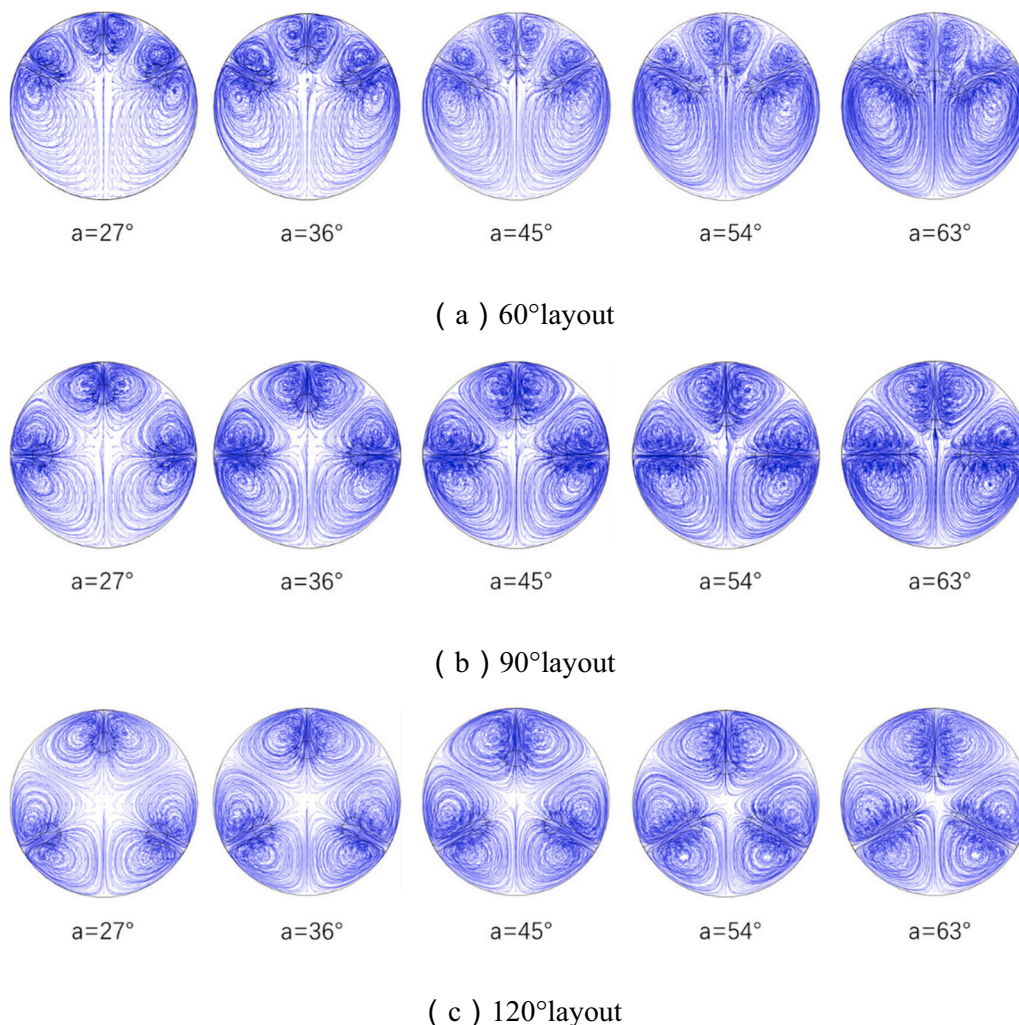


Fig. 5. Characteristics of the internal flow field in the reactor.

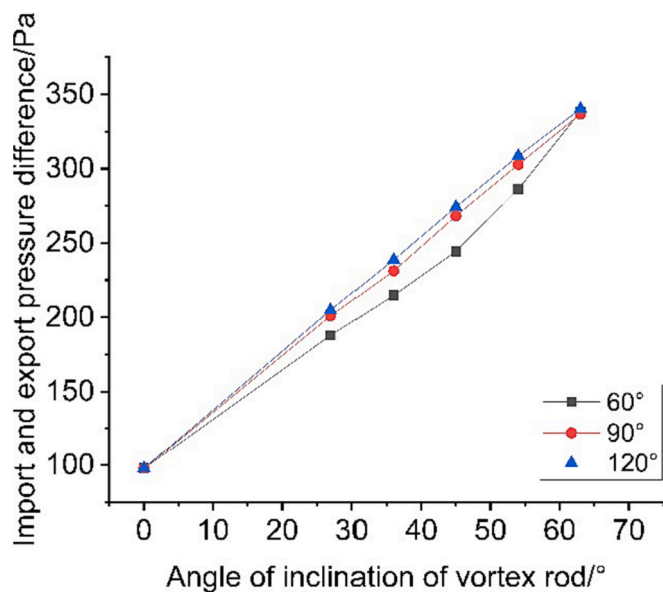


Fig. 6. The relationship between the inlet and outlet pressure difference of the reactor and the arrangement and the inclination angle of the vortex rod.

frequencies, but $\alpha = 45^\circ$ corresponds to a higher light-dark enhancement efficiency. Therefore, $\alpha = 45^\circ$ is a better choice for the 90° and 120° arrangements.

In Fig. 7(b), under a light field of $800 \mu\text{mol}/(\text{m}^2\cdot\text{s})$, the average light-dark cycle frequency in the photobioreactor with a 60° arrangement increases with an increasing inclination angle. The maximum frequency value in the simulation results ($f_{av} = 0.8945 \text{ Hz}$) is achieved when $\alpha = 63^\circ$. For the 90° arrangement with $\alpha = 45^\circ$, the peak average light-dark cycle frequency ($f_{av} = 0.80707 \text{ Hz}$) is obtained. Although the difference is not significant compared to $\alpha = 54^\circ$, considering the light-dark enhancement efficiency, $\alpha = 45^\circ$ is a better choice. When the vortex rod is arranged at 120° , the average light-dark cycle frequency reaches its maximum ($f_{av} = 0.71454$) when $\alpha = 54^\circ$.

In Fig. 7(c), under the irradiation of a light field of $1200 \mu\text{mol}/(\text{m}^2\cdot\text{s})$, the increase in the inclination angle of the photobioreactor with a 60° arrangement has the greatest effect on the average light-dark cycle frequency. The maximum values of f_{av} and μ are achieved when $\alpha = 63^\circ$, resulting in the best performance. For the 90° arrangement, $\alpha = 45^\circ$ and 54° have similar f_{av} values, but considering the overall light and dark enhancement efficiency, $\alpha = 45^\circ$ is a better choice. The maximum f_{av} value of $\alpha = 54^\circ$ is 0.68863 Hz when the vortex rod is arranged at 120° .

4.4. Effect of angle change on synergy angle

The addition of the vortex rod static mixer aims to fulfill the longitudinal vortex theory and enhance the utilization of light energy by

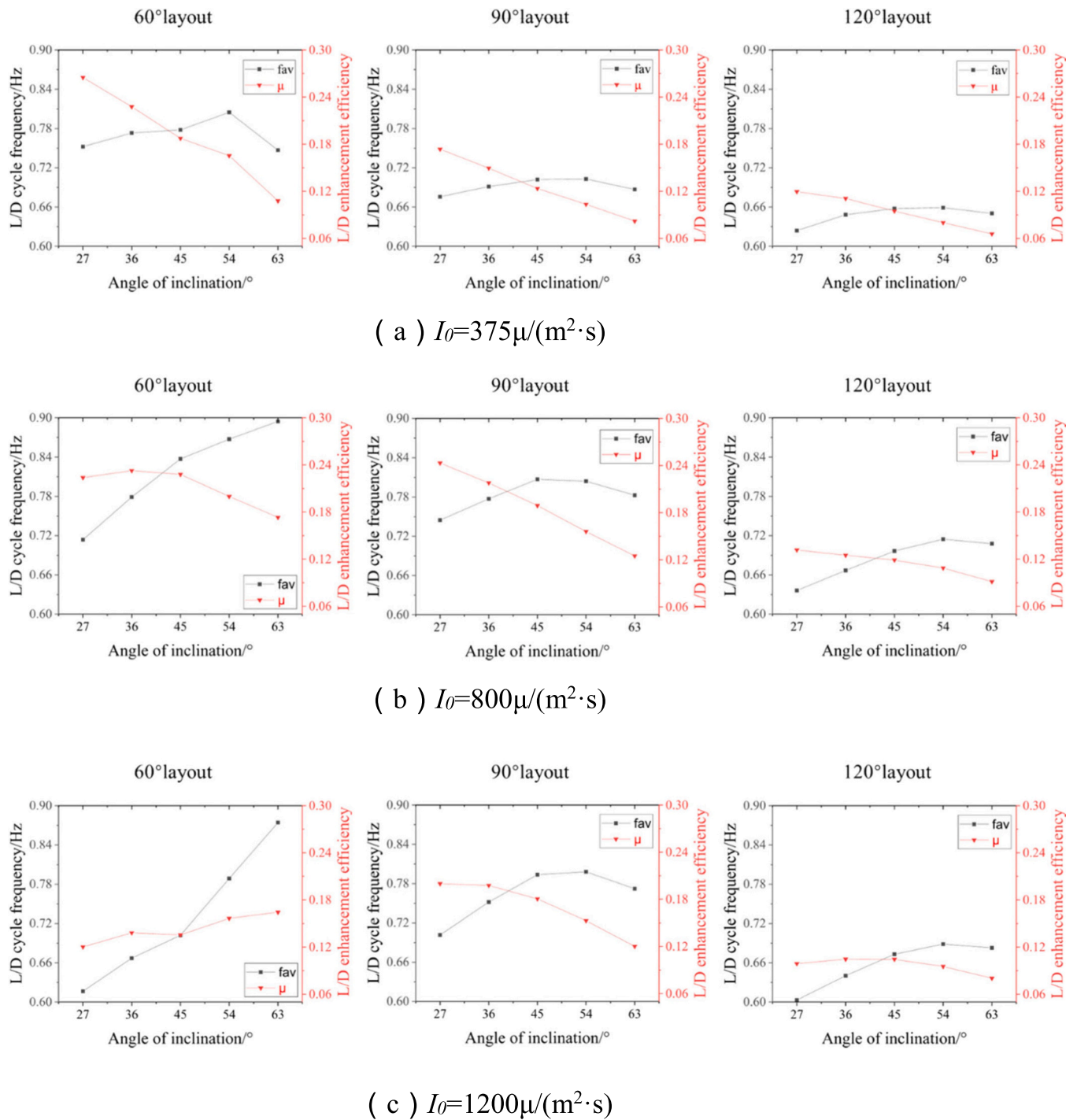


Fig. 7. The average light-dark cycle frequency and light-dark enhancement efficiency of the reactor under different illumination conditions vary with the inclination angle of the vortex rod.

microalgae. This involves generating vortices to improve the flow of the algae liquid in the direction of the light gradient. To assess the synergy between the light field and the velocity field, a representative vortex center area on one side of the symmetrical reactor tube can be chosen to study the absolute value of the cosine of the synergy angle in the light direction. In this study, with $x = 12.5$ mm, the results are presented in Fig. 8. The best synergy effect is observed when the synergy angle (β) is 0° or 180° , indicating that the closer $|\cos\beta|$ is to 1, the better the effect. Comparing Fig. 8(a), (b), and (c) with the smooth tube in Fig. 8(d), a more distinct and stable section of synergistic enhancement can be observed in the reactor after adding the vortex rod static mixer,

confirming the significance of the synergistic strengthening provided by the vortex rod. It is important to note that while Fig. 8(d) displays a considerable absolute value of the cosine of the synergy angle in the smooth tube, this is mainly due to the turbulent velocities in the radial direction of the algae solution. These random radial disturbance velocities are relatively small compared to the axial velocity, and the angle between the light field and the flow field is closer to 90° . Thus, the synergy effect in the smooth tube can be considered close to 0.

From Fig. 8(a), it can be observed that the $|\cos\beta|$ value of the algae liquid at the light-dark zone boundary is small when the arrangement is 60° , and the synergistic effect in the tube is mainly concentrated in the

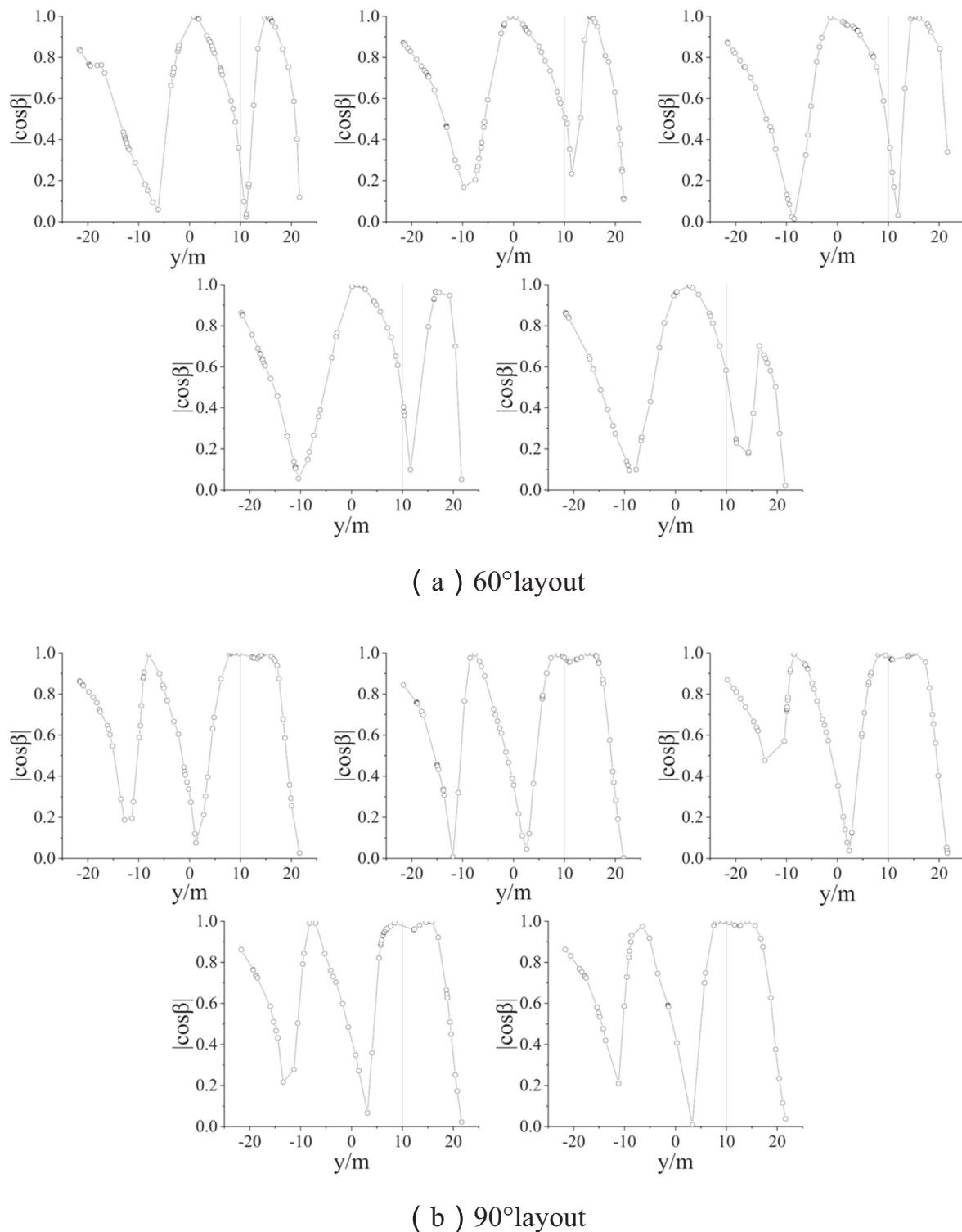
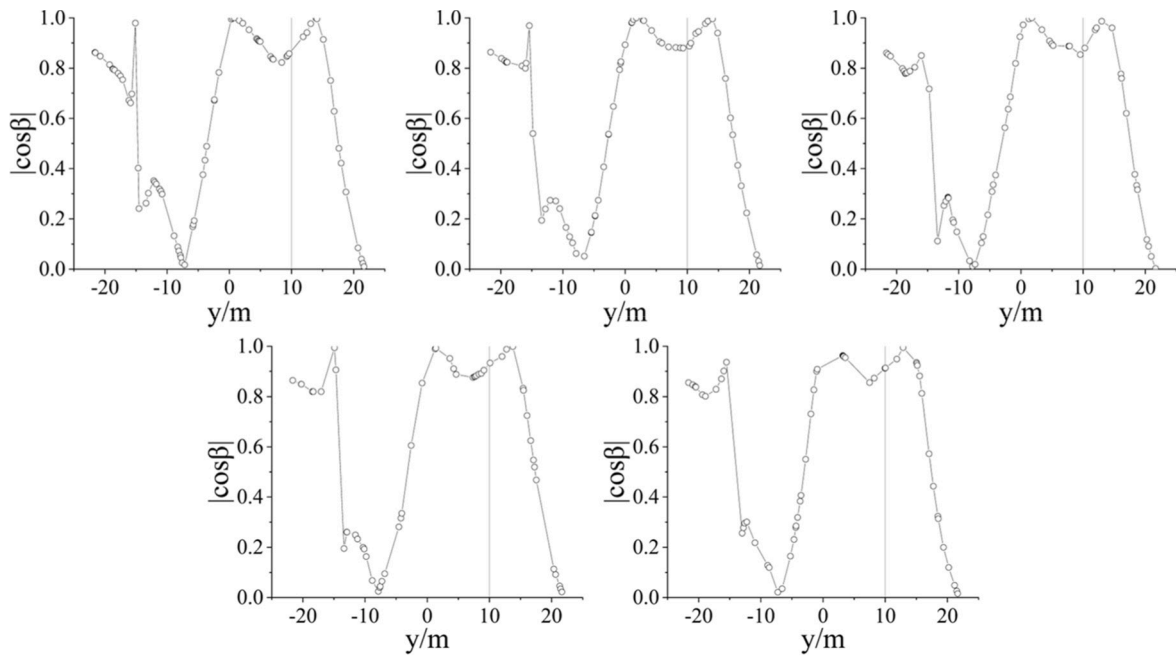


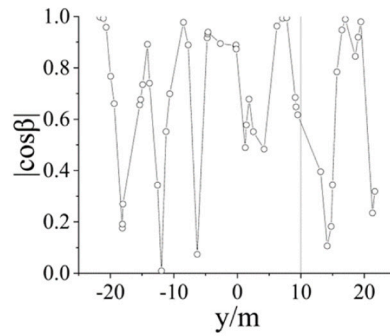
Fig. 8. The absolute value of the cosine of the synergy angle under different arrangements and inclination angles ($x = 12.5$ mm, $z = 1000$ mm).

light zone and the dark zone. Fig. 8(e) shows that the average $|\cos\beta|$ is highest in the light zone with an average value of 0.67475, and in the dark zone with an average value of 0.68635 when the vortex rod inclination angle (α) is 45° . In Fig. 8(b), it can be seen that $|\cos\beta|$ remains larger in the light-dark boundary area when the inclination angle is 90° . Further analysis from Fig. 8(e) reveals that when the inclination angle is above 45° , the dark area demonstrates a greater synergy effect.

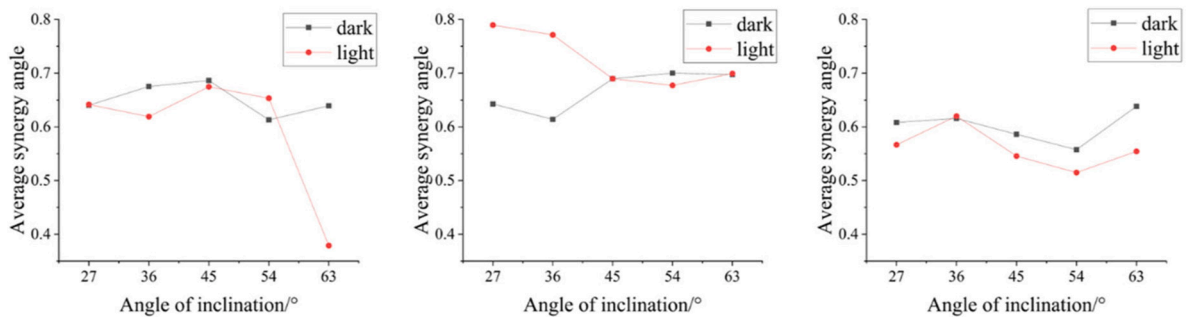
Conversely, when the inclination angle is less than 45° , the bright area exhibits a better synergy effect. When $\alpha = 45^\circ$, the overall synergy effect is balanced. Moving on to Fig. 8(c) shows that when the arrangement is 120° , $|\cos\beta|$ is also higher in the light-dark boundary area with only a slight attenuation at the boundary. There is also a stable area of $|\cos\beta|$ at the bottom of the dark area. Fig. 8(e) further demonstrates that the vortex rod has an advantage in enhancing the synergy effect in the dark



(c) 120°layout



(d) tube



(e) The average synergy angle of light and dark areas under each arrangement

Fig. 8. (continued).

area when the inclination angle is large. Specifically, when $\alpha = 63^\circ$, the average $|\cos\beta|$ in the dark area is 0.63832, while in the bright area, it is 0.55441. On the other hand, when $\alpha = 27^\circ$, the average $|\cos\beta|$ in the bright area reaches its maximum at 0.61991, which is higher than the

average $|\cos\beta|$ of 0.61573 at that point.

5. Conclusion

In this study, the focus is on investigating the enhancement efficiency of a photobioreactor with a vortex rod static mixer. The objective is to analyze the growth of algae solutions under different arrangements of the light tube and three types of vortex rods: 60°, 90°, and 120°, with varying inclination angles of the vortex rod (α) at 27°, 36°, 45°, 54°, and 63°. Through simulation using CFD software, parameters such as the average light-dark cycle frequency, the enhancement efficiency of the light-dark cycle, and the synergistic effect of the light field and velocity field were considered to obtain the following findings.

Under three illumination conditions (incident light fields $I_0 = 375 \mu\text{mol}/(\text{m}^2\cdot\text{s})$, $800 \mu\text{mol}/(\text{m}^2\cdot\text{s})$, and $1200 \mu\text{mol}/(\text{m}^2\cdot\text{s})$), an increase in the inclination angle (α) strengthens the vortex field and reduces the flow 'dead zone'. However, excessively large inclination angles can result in vortex deformation or even disappearance. The average light-dark cycle frequency initially increases and then decreases with the inclination angle, while the light-dark cycle enhancement efficiency shows a decreasing trend due to increased pressure drop.

Increasing light intensity leads to a gradual upward trend in the average light-dark cycle frequency and light-dark cycle enhancement efficiency in the reactor arranged with a 60° vortex rod. Examining the synergy between the illumination field and velocity field, it was found that for a 60° arrangement, the best synergy effect is achieved when the inclination angle (α) is 45°. In the case of a 90° arrangement, an inclination angle (α) greater than 45° is considered effective to ensure a balanced synergistic effect between the dark and light areas. For a 120° arrangement, the optimal cooperative arrangement is achieved with an inclination angle (α) of 36°, resulting in an average $|\cos\beta|$ value of 0.61782.

Based on these findings, it can be concluded that the optimal inclination angle for the vortex rod is 45°, and the arrangement angle should increase with the intensity of the incident light field. When the incident light field is $I_0 = 375 \mu\text{mol}/(\text{m}^2\cdot\text{s})$ and $800 \mu\text{mol}/(\text{m}^2\cdot\text{s})$, a 60° arrangement of the vortex rod is optimal. The average light-dark cycle frequency is 54.491 % and 66.258 % higher compared to the smooth tube, with an enhancement efficiency of 18.747 % and 22.795 %, respectively, and an average $|\cos\beta|$ value of 0.68055. When the incident light field is $I_0 = 1200 \mu\text{mol}/(\text{m}^2\cdot\text{s})$, a 90° arrangement of the vortex rod is recommended. The average light-dark cycle frequency is 57.580 % higher than that of the smooth tube, with an enhancement efficiency of 18.058 %, and an average $|\cos\beta|$ value of 0.68975.

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CRediT authorship contribution statement

Siding Li: Conceptualization, Methodology, Software, Investigation, Formal analysis, Writing – original draft. **Hua Wen:** Data curation, Writing – original draft. **Chuyi Peng:** Visualization, Investigation. **Chunyang Ma:** Conceptualization, Funding acquisition, Resources, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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